PAVEL YAKOVLEV

yvpavel@gmail.com · github.com/paulyakovlev

EXPERIENCE

General Motors April 2026 - Present
Software Engineer, Data Governance *Austin, TX*

- Built React and Django internal web application that automates Databricks onboarding — users submit forms and the application executes Databricks API calls to provision workspace access, create Azure AD groups, and manage group membership
- Administer Databricks platform across multiple workspaces, including user access provisioning, service principals, catalog permissions, external volumes, and Azure storage integration

General Motors April 2025 - April 2026
Data Engineer, Connected Vehicle Data Engineering *Austin, TX*

- Designed and deployed event-driven data collection systems, processing real-time telemetry across GM's connected vehicle fleet
- Resolved production data collection failures for rental fleet partner (100,000+ vehicles) by implementing direct read method to bypass queue bottleneck, ensuring accurate fuel level and odometer telemetry
- Developed Python ETL pipeline migrating CSV/XML configuration files from Git repository into Databricks, enabling PowerBI analytics for deployed data collection programs

General Motors February 2021 - April 2025
Software Engineer, Big Data Engineering & Infrastructure *Austin, TX*

- Developed GM's custom Hadoop distribution (GMDP) by integrating open-source components including Tez UI, Ambari, and Livy, migrating from Cloudera and reducing annual costs by $1.2M
- Contributed to Apache Ambari and Bigtop open-source projects by implementing Livy service integration (Ambari PR#3790, Bigtop PR#1282)
- Automated Docker-based Hadoop development environment, reducing developer setup time from 35+ minutes to 10 minutes by eliminating manual build and deployment steps
- Responded to Log4Shell vulnerability by developing automated remediation script for production Hadoop clusters, integrating into standard startup process for ongoing protection

PROJECTS

Coaster: E-Ink Cycling GPS

- Building ESP32-S3 cycling computer in C++ (PlatformIO) with a 200x200 e-paper display that records GPS and barometer data to GPX files and uploads rides to Strava over OAuth when it reaches home Wi-Fi
- Wrote a desktop simulator that renders every screen and a scripted ride through the real UI code, plus native unit tests for GPS parsing, ride stats and GPX output, so firmware is checked before flashing hardware

Real-Time Transit Data Pipeline

- End-to-end data pipeline collecting San Francisco Muni predictions via REST API, storing in PostgreSQL, and visualizing real-time transit patterns through Grafana dashboards
- Deployed on a self-hosted Raspberry Pi cluster, managing real hardware networking and node configuration to run Kubernetes workloads end-to-end

EDUCATION

University of California, Santa Cruz 2020
Bachelor's Degree in Computer Science

SKILLS

Languages: Python, Java, C++, Bash, SQL, JavaScript

Technologies: Node.js, React, Django, Docker, Kubernetes, PostgreSQL, Databricks, Azure, CI/CD